MATH 152
Mrs. Bonny Tighe

QUIZ 7

25 points
11.1 – 11.3

NAME Solutions

TA _____ Wed 10/22/2018

1. a. State the Geometric Series Test

2pts

$\sum_{n=1}^{\infty} a_n (r)^{n-1}$ is convergent if $|r| < 1$ and divergent if $|r| \geq 1$

b. State the test for Divergence

2pts

If $\lim_{n\to\infty} a_n \neq 0$ then $\sum_{n=1}^{\infty} a_n$ is divergent

c. State the Integral Test If f is positive, continuous and decreasing on $[1, \infty)$ and let $a_n = f(n)$ then the series $\sum_{n=1}^{\infty} a_n$ is

2pts

convergent if and only if the improper integral $\int_a^{\infty} f(x)\,dx$ is convergent

d. State the p-series test

1 pts

$\sum_{n=0}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$ and divergent if $p \leq 1$

2. Fill in the blank: a) If $\sum_{k=1}^{\infty} a_k$ *converges, then* $\lim_{k\to\infty} a_k =$ 0

b) What is the symbol for the 20th partial sum of a series? S_{20}

c) If $\sum_{k=1}^{\infty} a_k$ *diverges and* $\sum_{k=1}^{\infty} b_k$ *converges then* $\sum_{k=1}^{\infty} (ca_k + b_k)$ Diverges

d) If $\lim_{k\to\infty} a_k = 0$ *then* $\sum_{k=1}^{\infty} a_k$ converges. True or False False

$\frac{1}{2}$ point each

3. Find the limit of each sequence and state if the sequence converges or diverges. Is the sequence bounded? If so, what are the bounds. Determine whether the sequence is increasing, decreasing or not monotonic

a) $a_n = \dfrac{2^n}{n^3}$ 2pts

$\lim_{n\to\infty} \dfrac{2^n}{n^3} = \infty$

So divergent

$\dfrac{2}{1}, \dfrac{4}{8}, \dfrac{8}{27}, \dfrac{16}{81}, \dfrac{2^{10}}{10^3}$

$n=1$ $n=2$ $n=3$ $n=4$ $n=10$

not monotonic

not bounded

b) $a_n = \dfrac{3n^2+2}{5n^2+3}$ 2pts

$\lim_{n\to\infty} \dfrac{3n^2+2}{5n^2+3} = \dfrac{3}{5}$

So convergent

$\dfrac{5}{8}, \dfrac{14}{23}, \dfrac{29}{48}$

monotonic decreasing

bounded – upper bound 5/8

lower bound 3/5

4. Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^2+4k+3}$ 2pts

$\dfrac{A}{k+1} + \dfrac{B}{k+3} = \dfrac{1}{(k+3)(k+1)}$

$A(k+3) + B(k+1) = 1$

$k=-1$ $A = 1/2$

$k=-3$ $B = -1/2$

$\displaystyle\sum_{k=1}^{\infty}\left(\frac{1/2}{k+1} - \frac{1/2}{k+3}\right)$

$\dfrac{1/2}{2} + \dfrac{1/2}{3} + \dfrac{1/2}{4} + \dfrac{1/2}{5} + \dfrac{1/2}{6} \cdots$

$-\dfrac{1/2}{4} - \dfrac{1/2}{5} - \dfrac{1/2}{6}$

$\dfrac{1}{4} + \dfrac{1}{6} = \dfrac{5}{12}$ is the sum

So convergent

2pts b) $\displaystyle\sum_{k=0}^{\infty} \frac{2^{k+1}}{5^{k+2}}$ $\dfrac{2 \cdot 2^k}{5^2 \cdot 5^k} = \dfrac{2}{25}\left(\dfrac{2}{5}\right)^k$

$\displaystyle\sum_{k=0}^{\infty} ar^n$ convergent geometric series

$|r| = 2/5 < 1$

So Sum $= \dfrac{a}{1-r} = \dfrac{2/25}{1-2/5}$

$\dfrac{2}{25-10} = \dfrac{2}{15}$

5. Find the series. $1+\frac{1\cdot 4}{2!}+\frac{1\cdot 4\cdot 7}{4!}+\frac{1\cdot 4\cdot 7\cdot 10}{6!}+\ldots$

$n=1 \quad n=2 \quad n=3$

$$\sum_{n=1}^{\infty}\frac{1\cdot 4\cdot 7\cdots(3n-2)}{(2n-2)!} \quad \text{or} \quad \sum_{n=0}^{\infty}\frac{(1\cdot 4\cdot 7\cdots(3n+1))}{(2n)!}$$

1pt

6. Express $0.6\overline{5}$ as a ratio of integers using infinite series.

$$\frac{6}{10}+\sum_{n=0}^{\infty}\frac{5}{100}\left(\frac{1}{10}\right)^n=\frac{6}{10}+\frac{5/100}{1-1/10}=\frac{6}{10}+\frac{5}{100-10}=\frac{6}{10}+\frac{5}{90}$$

$$\frac{59}{90}$$

1pt

$r=\frac{1}{10}<1$
so Convergent

7. a) Estimate the error in using the partial sum S_{10} of the series $\sum_{k=1}^{\infty}\frac{1}{k^5}$

$$\lim_{t\to\infty}\int_{10}^{t}k^{-5}\,dk=\lim_{t\to\infty}-\frac{1}{4k^4}\Big]_{10}^{t}=\lim_{t\to\infty}\left[-\frac{1}{4t^4}+\frac{1}{4(10)^4}\right]$$

$$R_{10}\le\frac{1}{4(10)^4}$$

1pt

$$\boxed{\frac{1}{4(11)^4}\le R_{10}\le\frac{1}{4(10)^4}}$$

b) What partial sum would be necessary for the above series to have 10^{-4} accuracy?

$$\frac{1}{4(n)^4}\le\frac{1}{10^4}$$

1pt

$$10^4\le 4n^4$$

$$\frac{10}{\sqrt[4]{4}}\le n \text{ the next integer}$$

MATH 152
Mrs. Bonny Tighe

QUIZ 7A

25 points
11.1 – 11.3

NAME Solutions

TA _____ Wed 10/22/2018

1. a. State the Geometric Series Test

$\sum_{n=0}^{\infty} ar^n$ is convergent if $|r|<1$ and divergent if $|r| \geq 1$

b. State the test for Divergence

If $\lim_{n\to\infty} a_n \neq 0$ then $\sum_{n=1}^{\infty} a_n$ is divergent

c. State the Integral Test

If f is continuous, positive & decreasing on $[1,\infty)$ and let $a_n = f(n)$ then $\sum_{n=1}^{\infty} a_n$ is convergent if and only if $\int_1^{\infty} f(x)dx$ is convergent.

d. State the p-series test

$\sum_{n=1}^{\infty} \frac{1}{n^p}$ is Convergent if $p>1$ and is divergent if $p \leq 1$

2. Fill in the blank: a) If $\sum_{k=1}^{\infty} a_k$ *converges, then* $\lim_{k\to\infty} a_k =$ 0

b) What is the symbol for the 30th partial sum of a series? S_{30}

c) If $\sum_{k=1}^{\infty} a_k$ *diverges and* $\sum_{k=1}^{\infty} b_k$ *converges then* $\sum_{k=1}^{\infty}(ca_k + db_k)$ Diverges

d) If $\lim_{k\to\infty} a_k = 0$ *then* $\sum_{k=1}^{\infty} a_k$ converges. True or False False

3. Find the limits of the sequences and state if the sequence converges or diverges. Is the sequence bounded? If so, what are the bounds. Determine whether the sequence is increasing, decreasing or not monotonic

a) $a_n = \frac{n^3}{3^n}$

$\lim_{n \to \infty} \frac{n^3}{3^n} = 0$ so convergent

$\frac{1}{3}, \frac{8}{9}, \frac{27}{27}, \frac{64}{81}$

not monotonic

Bounded - upper bound $= 1$

lower bound $= 0$

b) $a_n = \frac{2n+1}{3n+2}$

$\lim_{n \to \infty} \frac{2n+1}{3n+2} = 2/3$ so convergent

$\frac{3}{5}, \frac{5}{8}, \frac{7}{11}, \frac{9}{14}$

monotonic increasing

bounded - lower bound $= 3/5$

upper bound $= 2/3$

4. Determine whether the series is convergent or divergent and if it is convergent **find the sum.**

a) $\sum_{k=1}^{\infty} \frac{1}{k^2 + 4k + 3}$

$\frac{A}{k+1} + \frac{B}{k+3} = \frac{1}{(k+1)(k+3)}$

$A(k+3) + B(k+1) = 1$

$k = -1 \quad A = 1/2$

$k = -3 \quad B = -1/2$

$\sum_{k=1}^{\infty} \left(\frac{1/2}{k+1} - \frac{1/2}{k+3} \right)$

$\frac{\frac{1}{2}}{2} + \frac{\frac{1}{2}}{3} + \frac{\frac{1}{2}}{4} + \frac{\frac{1}{2}}{5} + \frac{\frac{1}{2}}{6} +$

$- \frac{1/2}{4} - \frac{1/2}{5} - \frac{1/2}{6} -$

Sum $= \frac{1}{4} + \frac{1}{6}$

I found the sum so the series is convergent

b) $\sum_{k=1}^{\infty} \frac{e^{k+1}}{5^{k+2}}$

$\boxed{\sum_{k=1}^{\infty} ar^{k-1}}$

$\sum_{k=1}^{\infty} \frac{e^2 \cdot e^{k-1}}{5^3 \cdot 5^{k-1}} = \frac{e^2}{125} \left(\frac{e}{5}\right)^{k-1}$

$|r| = e/5 < 1$ so convergent geometric series

Sum $= \frac{\frac{e^2}{125}}{1 - e/5}$

$= \frac{e^2}{125 - 25e}$

8. Use the **Integral test** to determine if the series is divergent or convergent.

3pts

$$\sum_{n=0}^{\infty} n^2 e^{-n^3}$$

$f(n) = \frac{n^2}{e^{n^3}}$ which is positive, decreasing + continuous for $n \geq 0$

$$\lim_{t\to\infty} \int_0^t n^2 e^{-n^3}\,dn =$$

$$\boxed{\begin{aligned} u &= -n^3 \\ du &= -3n^2\,dn \end{aligned}}$$

$$\lim_{t\to\infty} \left[-\frac{1}{3}e^{-n^3}\right]_0^t = \lim_{t\to\infty} \left[-\frac{1}{3e^{t^3}} - \left(-\frac{1}{3e^0}\right)\right] = \frac{1}{3}$$

So Convergent

Thus $\sum_{n=0}^{\infty} n^2 e^{-n^3}$ is also convergent

$n=1 \quad n=2 \quad n=3 \quad n=4$

5. Find the series. $1+\frac{1\cdot 3}{2!}+\frac{1\cdot 3\cdot 5}{3!}+\frac{1\cdot 3\cdot 5\cdot 7}{4!}+\ldots$

$$\sum_{n=1}^{\infty}\frac{(1\cdot 3\cdot 5\cdots(2n-1))}{n!} \quad \text{or} \quad \sum_{n=0}^{\infty}\frac{(1\cdot 3\cdot 5\cdots(2n+1))}{(n+1)!}$$

6. Express $0\,0.4\overline{3}$ as a ratio of integers using infinite series.

$$\frac{4}{10}+\sum_{n=0}^{\infty}\frac{13}{100}\left(\frac{1}{10}\right)^n = \frac{13/100}{1-1/10} = \frac{4}{10}+\frac{13}{100\cdot 10} = \frac{4}{10}+\frac{3}{90}$$

$$\frac{39}{90}$$

$r=\frac{1}{10}<1$

so convergent

7. Use the **Integral test** to determine if the series is divergent or convergent.

$\sum_{n=1}^{\infty} ne^{-n^2}$ let, $f(n)=\frac{n}{e^{n^2}}$ is continuous, positive + decreasing on $[1,\infty)$

$$\lim_{t\to\infty}\int_1^t ne^{-n^2}\,dn \qquad \boxed{\begin{aligned} u &= -n^2 \\ du &= -2n\,dn \end{aligned}}$$

$$\lim_{t\to\infty} -\frac{1}{2}e^{-n^2}\Big]_1^t = \lim_{t\to\infty}\left[-\frac{1}{2e^{t^2}}+\frac{1}{2e^1}\right] = 0+\frac{1}{2e}=1/2e$$

So convergent

$\therefore \sum_{n=1}^{\infty} ne^{-n^2}$ is also convergent

$$\int_{n+1}^{\infty} f(n)\,dn \le R_n \le \int_{n}^{\infty} f(n)\,dn$$

8. a) Estimate the error in using the partial sum S_{10} of the series $\sum_{k=1}^{\infty} \frac{1}{k^3}$

$$\lim_{t\to\infty} \int_{10}^{t} k^{-3}\,dk = \lim_{t\to\infty} \left. -\frac{1}{2k^2} \right]_{10}^{t} = \lim_{t\to\infty} \left[-\frac{1}{2t^2} + \frac{1}{2(10)^2} \right]$$

$$\frac{1}{2(11)^2} \le R_{10} \le \frac{1}{2(10)^2}$$

b) What partial sum would be necessary for the above series to have 10^{-4} accuracy?

$$R_{10} \le \frac{1}{2(10)^2}$$

$$\frac{1}{2(n^2)} \le \frac{1}{10^4}$$

$$10^4 \le 2n^2$$

$$\frac{100}{\sqrt{2}} \le n \text{ the next integer}$$